No-Distance Combinatorial Austerity Line from Metric Removal to Harmonic-Address Selection, Armed-State Irreversibility, C0-to-DK Bridge Tests, and Integrated Austerity Resolution

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Abstract

This paper updates the HAOS-IIP combinatorial austerity line from Stage C0.1 through Stage C0.20 and adds an integrated Stage 11 convergence probe. The line begins as a strict no-distance control: coordinate distance is removed from the interaction kernel and replaced by graph-internal structure on an explicit edge-graph control branch. It then expands in four directions. The first direction, Stages C0.5–C0.9, asks whether combinatorial survivor rules, incidence-seeded mismatch accumulation, and softer bounded-path kernels can produce sequencing, explicit kernel inconsistency, or broad irreversibility. The second direction, Stages C0.10–C0.16, adds a static harmonic-address compatibility factor on the clustered Dirac–Kahler seed and asks whether topology selection sharpens into a resonance ladder, a detuning phase boundary, hysteretic return error, selector-withdrawal retention, or a locking mechanism. The third direction tests whether a signed combinatorial Dirac–Kahler candidate can be constructed directly from C0 primitives and whether any part of the harmonic detuning and protection structure survives on that derived complex. The fourth direction starts from that bridge scaffold and asks the fairest late questions: does a recovered bridge braid window survive selector withdrawal, does it leave an armed-state hysteresis witness, and do the open bridge obligations converge when mismatch orientation, signed evolution, harmonic-address sectoring, and topology survival are measured synchronously. The combined result is bounded and mixed. Graph-structural class survival survives metric removal, and incidence-forced protected sequencing appears reproducibly under external combinatorial forcing. A weaker bounded-path survivor kernel can open explicit adjacency-versus-kernel inconsistency in five of nine runs, but that partial positive does not scale under tri-root forcing. The harmonic-address branch shows a weak but real topology-selection signal: exact-match sectors preserve the braid class, resonance alignment stratifies into address bands, and a braid-to-smear transition appears near detuning $\delta \approx 0.75$. The bridge validation closes the signed chain identities exactly on the tested representative graph complexes, and a corrected armed-state bridge hysteresis probe later opens explicit irreversibility witnesses on the clustered family. But fair bridge withdrawal still fails to retain topology, and the integrated Stage 11 probe resolves negatively overall: orientation and evolution couple cleanly, while intrinsic grading and family-wide recovery remain unconverged. The updated boundary is therefore sharper than before: the C0 line supports graph-structural class survival, limited protected sequencing, weak harmonic-address topology selection, clustered armed-state irreversibility on the bridge scaffold, and a real C0-to-DK bridge candidate, but it still does not support topology locking or a family-wide structuralization of harmonic address on the no-distance branch.

1 Introduction

The C0 line is a bounded control program inside HAOS-IIP, not a replacement core. Its purpose is narrower:

> *What survives once coordinate distance is removed, and what further combinatorial ingredients are still insufficient to turn selection into irreversibility or locking?*

The early answer was reported in Paper 24.1, which froze Stages C0.1 through C0.4 as a metric-removal and topology-control sequence. The present paper extends that line through C0.20 and a first integrated austerity-resolution probe while keeping the core/model boundary explicit:

- the frozen core remains the HAOS-IIP axioms of recoverability, interaction primacy, representational regimes, viability-constrained dynamics, and invariant residue;
- the C0 line remains a regime-specific control branch beneath that core;
- positive and negative numerical outcomes are treated as branch boundaries, not as revisions of the axioms.

2 Core Boundary and Spectral Framing

The recent canonical cleanup matters here. On Laplacian-style branches, HAOS recoverability can be written as an operator-level spectral condition rather than as free-floating language. If

$$L = D - A, \qquad 0 = \lambda_0 \leq \lambda_1 \leq \lambda_2 \leq \cdots ,$$

then the cleanest scalar-branch compression is

$$e^{-\lambda_1 T}\delta < \epsilon_c,$$

where δ is perturbation size, T is the admissible recovery horizon, and ϵ_c is the viable coherence threshold. In words: the low end of the interaction spectrum must damp perturbations back below the viable coherence band quickly enough to restore operational structure.

This sharpens the language of recoverability on Laplacian branches, but it does not rewrite any C0 outcome. It clarifies how the scalar control line should be read:

- C0 positives remain positives only if recoverability improves inside the branch;
- C0 nulls remain nulls even if a branch looks visually structured;
- a partial opening of kernel inconsistency is not the same thing as broad irreversibility;
- topology selection is not the same thing as topology locking.

3 Design Discipline Across the C0 Line

The C0 line actually contains two sub-branches that share a common discipline but answer different questions.

3.1 C0.1–C0.9: pure no-distance control branch

These stages use an explicit projected-transverse edge-graph control operator. The graph size, representative encounter families, and shell-weight measurement vocabulary are fixed while the kernel or survivor rule is varied using only graph-internal quantities:

- adjacency,
- bounded path length in edge count,
- local degree,
- local motif incidence,
- combinatorial Laplacian structure.

The three representative families are:

- `clustered_composite_anchor`,
- `counter_propagating_corridor`,
- `phase_ordered_symmetric_triad`.

3.2 C0.10–C0.14: harmonic-address extension

These stages shift to the clustered Phase III Dirac–Kahler seed while keeping the projected-transverse sector, graph resolution, and topology measurement stack fixed. A static harmonic-address compatibility factor is added as a relational multiplier on the combinatorial operator. The branch therefore asks a different question:

> *Can harmonic-address compatibility select, stratify, or lock topology without metric distance?*

4 Branch A: Metric Removal and Topology Controls (C0.1–C0.4)

Paper 24.1 already froze the first four stages. The present synthesis keeps that result intact:

- C0.1: primary persistence-class behavior survives removal of coordinate-distance weighting on the explicit edge-graph control branch, although transport texture shifts.
- C0.2: degree-spectrum variation reshapes transport span and flow concentration but does not overturn the coarse topology labels.
- C0.3: the no-distance branch does not produce a clean braid baseline; the nearest honest surrogate, the counter-propagating corridor, remains `smeared_transfer` under all tested incidence-noise perturbations.
- C0.4: small local motifs do not seed a reproducible new topology class; eight of nine runs are neutral and the only non-neutral response is an isolated diamond-induced channel-splitting case.

The stable conclusion from C0.1–C0.4 therefore remains:

> *The no-distance branch supports graph-structural class survival and transport reshaping, but not a reproducible combinatorial braid family or a motif-seeded topology mechanism.*

5 Branch B: Sequencing, Mismatch, and Kernel Inconsistency (C0.5–C0.9)

5.1 C0.5: static sequencing baseline

Stage C0.5 replaces the assumed packet-evolution clock with a combinatorial sequencing rule built from persistence-class stabilization, incidence-mismatch accounting, kernel-update consistency, and protected topology-stability blocks. The result is mostly static. Eight of nine runs freeze into `static_regime`, the mismatch ledger stays zero everywhere, and only one protected ordering event appears on the symmetric triad.

This is a useful negative baseline: the unforced branch is too rigid to generate sequencing internally.

5.2 C0.6: incidence-seeded mismatch accumulation

Stage C0.6 deliberately injects the noise and motif menus from C0.3 and C0.4 after the protected stability constraint is activated. Here the branch opens clearly:

- positive mismatch density appears in 9/9 runs,
- protected arrow forcing appears in 6/9 runs,
- the protected family labels remain intact: three `trapped_local`, three `smeared_transfer`, three `split_channel_exchange`.

This is the strongest positive result in the sequencing line. The branch does support incidence-forced protected ordering. But the mechanism is still not inconsistency-mediated because kernel-consistency failures remain zero.

5.3 C0.7: original alternation remains too rigid

Stage C0.7 adds explicit adjacency-versus-graph-shell alternation on top of the positive-mismatch runs from C0.6. The result is cleanly negative:

- explicit consistency failures remain 0/9,
- protected arrow under inconsistency remains 0/9,
- the general protected ordering inherited from C0.6 survives only as an ordering-without-inconsistency effect.

So the original survivor-kernel pair is too rigid to support explicit inconsistency as the driver.

5.4 C0.8 and C0.9: softer survivor maps partially open, then fail to scale

Stage C0.8 replaces the rigid graph-shell survivor map with a bounded-path kernel weighted by $1/(h+1)$ up to a fixed hop cap and upgrades motifs to bi-root and two-layer injections. This is the first stage that opens explicit inconsistency:

- explicit consistency failures appear in 5/9 runs,
- protected-arrow events under weaker inconsistency appear in 2/9 runs,

Stage	Positive mismatch	Explicit inconsistency	Protected arrow	Main read
C0.5	0/9	0/9	1/9	mostly static sequencing baseline
C0.6	9/9	0/9	6/9	incidence-forced protected ordering
C0.7	9/9	0/9	0/9	original alternation too rigid
C0.8	9/9	5/9	2/9	partial inconsistency opening under weaker kernel
C0.9	9/9	4/9	0/9	tri-root scaling fails

Table 1: Sequencing and inconsistency block. The branch supports reproducible mismatch accumulation and a partial inconsistency opening under a softer bounded-path survivor map, but not broad irreversibility scaling.

- the hybrid noise-plus-motif protocol is the only forcing family that produces protected ordering under explicit inconsistency, with positive cases on the clustered anchor and the symmetric triad.

Stage C0.9 tries to scale that partial opening with tri-root and three-layer motifs. It fails:

- explicit consistency failures drop to 4/9,
- tri-root-specific failures appear in only 2/9,
- protected-arrow events under weaker or tri-root inconsistency fall to 0/9,
- the irreversibility scaling factor is 0.0 relative to the C0.8 partial positive.

The correct combined boundary for C0.5–C0.9 is therefore:

> *The no-distance branch supports incidence-forced protected sequencing and can be pushed into explicit inconsistency under softer survivor maps, but it does not support a broad claim that irreversibility follows automatically from that inconsistency.*

6 Branch C: Harmonic-Address Selection, Detuning, Hysteresis, and Selector Withdrawal (C0.10–C0.16)

6.1 C0.10: weak proto-selection, no persistence gain

Stage C0.10 adds a static harmonic-address compatibility factor to the clustered Dirac–Kahler seed. This opens the first clean topology-selection signal:

- exact-match runs preserve the braid class in 2/9 runs,
- near-match and incompatible sectors relax to `transfer_smeared`,
- persistence gain remains exactly 0.0 across the matrix.

So harmonic addressing acts as a weak proto-selection rule, not as a persistence generator.

6.2 C0.11 and C0.12: ladder plus threshold

Stage C0.11 asks whether discrete harmonic-address differences generate a resonance ladder. The answer is yes, but in a split sense:

- topology shows two visible bands: same-address versus all nonzero address differences,

Stage	Main class counts	Strongest positive	Boundary
C0.10	2 braid, 7 smear	exact-match address preserves braid	no persistence gain
C0.11	2 braid, 7 dispersive	resonance ladder in alignment	same-address braid not refinement-stable
C0.12	6 braid, 3 smear	sharp flip near $\delta = 0.750$	not a smooth decay and not multi-plateau
C0.13	6 protected braid, 36 smear	narrow local protected window	no family satisfies strict 3-sample rule
C0.14	3 braid, 6 smear	none	all 9/9 runs are `no_locking_effect`

Table 2: Harmonic-address block. Harmonic addressing selects and stratifies topology, but the branch still closes on a locking null result.

- resonance alignment further stratifies the nonzero sector into $d1/d4$ and $d2/d3$.

The same-address braid sector is not fully refinement-stable, so the ladder is stronger in alignment than in topology stability.

Stage C0.12 then scans a denser detuning ladder. The result is a sharp threshold:

- `braid_like_exchange` persists through $\delta = 0.625$ in 6/9 runs,
- the topology flips once near $\delta = 0.750$,
- the remaining three runs are `transfer_smeared` for $\delta = 0.750, 0.875, 1.000$.

6.3 C0.13 and C0.14: boundary refinement and locking null

Stage C0.13 resolves the detuning corridor near the observed transition. The local picture is sharp:

- in the baseline, motif-weak-minus, and beta-weak families, `address_protected_braid` survives at $\delta = 0.66, 0.69$ and flips to `transfer_smeared` at $\delta = 0.72$,
- motif-weak-plus, degree-mild, and degree-asymmetric collapse the protected window before the scan even enters the corridor.

The global verdict remains `fragmented multi-window protection` because no family satisfies the strict three-consecutive-sample resolution rule. The boundary is visually sharp in the baseline-like families but not formally resolved by the atlas criterion.

Stage C0.14 then tests the next honest escalation: can bounded local latches convert topology selection into locking? The answer is no.

- all 9/9 runs remain `no_locking_effect`,
- no `locked_braid` or `quasi_bounded_exchange` class appears,
- the exact-match family stays `braid_like_exchange`,
- the boundary and fragile families stay `transfer_smeared`,
- the harmonic phase latch actually shortens braid survival on the exact-match seed by 0.1253.

The correct combined boundary for C0.10–C0.14 is:

> *Harmonic addressing produces weak topology selection and a real detuning threshold on the clustered seed, but the tested bounded latch rules do not convert that selection into locking.*

6.4 C0.15 and C0.16: hysteresis and selector-withdrawal do not rescue locking

The next two stages ask the sharper derivational questions that C0.14 leaves open.

Stage C0.15 replaces one-way detuning scans with a forward/reverse hysteresis protocol. The same harmonic-address family is driven out and back through the detuning corridor while the return error and hysteresis area are measured directly. The result is a partial positive only:

- 4/9 runs classify as `irreversible_witness`,
- 5/9 runs classify as `hysteretic_retrace`,
- the strongest non-return cases are the exact plain, exact motif, fragile degree, and fragile motif protocols,
- but broad irreversibility is not established.

So the branch does support path dependence, but not a strong claim that irreversibility now follows generically from harmonic-address forcing.

Stage C0.16 then asks the cleaner locking question: once a topology has been selected, does it remain after the selector is removed? The answer is a clean no.

- 0/9 runs produce any retained locking class,
- 3/9 runs classify as `selector_dependent_collapse`,
- 6/9 runs classify as `no_locking_after_withdrawal`,
- the threshold-protected family can still arm into braid under selection, but immediately collapses when the selector is withdrawn.

These two stages sharpen the earlier locking null rather than overturning it. The honest extension of the harmonic-address boundary is therefore:

> *The harmonic-address branch supports weak topology selection, a real detuning threshold, and partial hysteresis, but selected topology does not survive selector withdrawal and therefore still does not qualify as locked.*

7 C0-to-DK Bridge Validation and Late Bridge Extensions

The bridge note is not a new persistence claim. It is a validation scaffold that asks whether a signed combinatorial Dirac–Kahler candidate can be built directly from C0 primitives and whether any part of the late C0 selection/protection structure survives on that derived complex.

The first important result is algebraic rather than phenomenological. On the three representative graph families

- `clustered_composite_anchor`,
- `counter_propagating_corridor`,
- `phase_ordered_symmetric_triad`,

the signed bridge complex closes exactly:

$$\|d_1 d_0\| = 0, \qquad \|D^2 - \Delta\| = 0, \qquad \|d_0^T d_0 - L_0\| = 0.$$

So the bridge candidate is not just a verbal analogy. It is a real signed chain complex on the tested graph families.

The second result is selective and narrower. On the derived complex:

- the clustered family opens a measurable braid-to-smear bridge corridor, with `braid_like_exchange` at $\delta = 0.0, 0.125$ and `transfer_smeared` from $\delta = 0.25$ onward;
- the focused clustered protection panel reproduces the full tested qualitative pattern: balanced baseline braid, phase-edge smear, width-asymmetry smear, degree-skew tolerance, and zero-form fragility;
- the corridor and triad families do not recover a broad braid sector.

The bridge note therefore earns a bounded but real statement:

> *A combinatorial Dirac–Kahler candidate can be constructed directly from C0 primitives, and its signed chain identities close exactly on the tested representative graph complexes. But orientation source, evolution-law equivalence, intrinsic grading source, and family-wide protection recovery remain conditional.*

This is useful because it tightens the HAOS-IIP core/model boundary rather than weakening it. The graded sector is no longer only an imported Phase III object; there is now a real C0-derived bridge scaffold. But it remains a scaffold, not a finished derivation.

7.1 C0.17–C0.20: fair withdrawal stays negative, armed-state hysteresis opens irreversibility

The first two bridge-local latch stages, C0.17 and C0.18, were useful but not yet fair enough as retention tests. Both were fully negative, but they mixed failure-to-arm with failure-to-retain: the clustered bridge family often left the braid class during arming itself.

Stage C0.19 corrects that by withdrawing from the real recovered clustered bridge braid window rather than from a fixed arming cutoff. The corrected read is stronger, not weaker:

- the clustered bridge family does recover a real braid window at steps 138–142,
- all three fair withdrawal protocols then collapse that armed state to `unresolved_mixed`,
- post-withdrawal braid dwell is only 1.0 step in every clustered run,
- the best label is `topology_afterglow`, not locking.

So the late locking null is no longer vulnerable to the objection that the bridge stages withdrew from the wrong part of the trajectory. Even when the selector is removed from a real bridge braid window, topology does not hold itself together.

Stage C0.20 then asks the irreversibility question from that same armed state. This is the strongest late positive now available on the bridge scaffold:

- all three clustered armed-state forward/reverse protocols finish in `transfer_smeared` even after exact-match restoration,

- all three clustered runs therefore classify as `irreversible_witness`,
- the strongest protocol is the frozen-hold cycle, with hysteresis areas 0.1341 in flow and 0.0977 in loop score,
- corridor and triad still fail to open armed braid windows on the bridge scaffold and therefore remain `no_armed_braid_window`.

The combined bridge extension is therefore asymmetric:

> *The bridge scaffold now supports explicit armed-state irreversibility on the clustered family, but even a fair selector-withdrawal test does not produce topology locking.*

8 Stage 11 Integrated Austerity Resolution

Stage 11 is the first convergence probe rather than a narrow micro-control. It forces four previously open structures to co-evolve synchronously on matched graph families:

- mismatch-ledger orientation,
- signed combinatorial field evolution,
- intrinsic harmonic-address sectoring,
- topology survival.

The point is to test whether any positive signal still survives once these structures are measured in the same state history rather than across separate runs.

The result is globally negative, but in a sharpened way:

- `O_CO_ORIENT` is positive: two families show persistent positive ledger alignment, with medians 0.9996 on the clustered seed and 0.9937 on the triad;
- `O_CO_EVOLUTION` is positive: the coarse/fine evolution mismatch ratio remains tightly bounded around 1.0056;
- `O_CO_GRADING` is negative: sector entropy drops only weakly, from 0.2114 to 0.2091, and does not resolve into a convincing intrinsic-grade structure;
- `O_CO_RECOVERY` is negative: the required ordering `clustered` $\geq$ `corridor` $\geq$ `triad` fails in every setting, so the ordered-setting fraction is 0.0000.

So Stage 11 does not collapse the bridge obligations into a single structural win. It splits them:

> *Mismatch-orientation and signed-field evolution now look genuinely coupled, but harmonic address still behaves more like selector phenomenology than intrinsic grading, and family-wide recovery remains unresolved.*

9 What Changes and What Does Not

The updated paper changes the map of the C0 line, but not the core or the main phase boundaries.

9.1 What changes

- The C0 line is no longer just a metric-removal control. It now includes a full sequencing/inconsistency block, a harmonic-address selection/locking block, a late bridge irreversibility block, and a synchronous integrated-resolution probe.
- The strongest late positive is no longer "metric removal preserves some classes." It is C0.6: incidence-forced protected sequencing with positive mismatch in 9/9 runs and protected arrow forcing in 6/9 runs.
- The strongest late negative is no longer just "motifs do not seed braid." It is now the combined C0.14–C0.19 boundary: selection does not become locking under bounded latches, selected topology does not survive selector withdrawal, and even fair bridge withdrawal from a real braid window yields only topology afterglow.
- A new bridge-level positive now exists at two levels: the C0-to-DK signed complex closes exactly on the tested representative graph families and recovers a clustered-only detuning/protection signal, and C0.20 later adds an explicit armed-state irreversibility witness on that same clustered bridge scaffold.
- Stage 11 changes the obligation map: orientation and evolution now look coupled enough to survive an integrated test, while grading and family-wide recovery still do not.

9.2 What does not change

- The frozen HAOS-IIP core axioms do not change.
- Paper 24.1 remains the frozen early synthesis for C0.1–C0.4.
- Phase I remains closed by the same boundary: frozen pre-geometry can organize, but it does not bind.
- Phase II is still not rescued by the C0 line. C0.8 opens only a partial inconsistency signal and C0.9 fails to scale it.
- Phase III topology findings remain topology findings; the harmonic-address branch adds a selector on that clustered seed but not a persistence law, and neither the bridge extensions nor Stage 11 collapse the gap between C0 and the full Phase III graded sector.

10 Conclusion

The full C0.1–C0.20 line, together with the integrated Stage 11 probe, now has a stable architecture and a cleaner final boundary.

First, metric removal does not erase the no-distance branch. Real graph-structural class survival remains after coordinate distance is removed. Degree landscape, incidence noise, and motif forcing reshape transport texture and local class behavior, but they do not magically produce a robust braid family.

Second, sequencing and inconsistency are not automatic. The branch supports reproducible mismatch accumulation and protected ordering under external forcing, and a weaker bounded-path survivor map can open explicit kernel inconsistency. But that opening is partial and does not scale into a broad irreversibility claim. Later hysteresis probes show path dependence in part of the

harmonic-address matrix, not a broad irreversible law, and the integrated Stage 11 run still does not turn that path dependence into family-wide recovery.

Third, harmonic addressing behaves like a selector, not a locker. Exact-match sectors preserve braid on the clustered seed, resonance alignment stratifies into discrete address bands, and a narrow detuning threshold appears near $\delta \approx 0.75$ on the original clustered DK branch. But bounded latches do not convert that selection into a locked or quasi-bounded exchange regime, selector withdrawal collapses the selected topology instead of retaining it, and even fair bridge withdrawal from a real recovered braid window yields only one-step afterglow.

Fourth, the graded bridge is now real but still conditional. A C0-derived signed Dirac–Kahler candidate closes exactly on the tested representative graph complexes and recovers a clustered-only detuning/protection signal. From that scaffold, armed-state bridge hysteresis now yields an explicit clustered irreversibility witness. But the integrated Stage 11 run shows that intrinsic grading and family-wide recovery still do not converge. That is enough to justify the bridge as a scaffold with a real late irreversibility signal, not enough to claim that the graded sector is fully derived from C0 alone.

The final C0 boundary can therefore be stated without hype:

> *The no-distance combinatorial line supports graph-structural class survival, limited protected sequencing, weak harmonic-address topology selection, explicit clustered armed-state irreversibility on the bridge scaffold, and a real C0-to-DK bridge candidate, but it does not yet support topology locking, intrinsic grading, or family-wide recovery on the no-distance branch.*

That is not a failure of the line. It is exactly the kind of bounded map a control program is supposed to produce.